

Real-Time Prediction of Online Shoppers' Purchasing Intention Using MLP and LSTM Recurrent Neural Networks

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Abstract

This work proposes a real-time behavior analysis system for online shoppers. The system has two connected modules: one estimates whether a visitor intends to purchase, and the other estimates whether that visitor is likely to leave the site soon. For purchase-intention prediction, the study uses aggregated page-view behavior, session attributes, and user-related information. These features are evaluated with random forests, support vector machines, and multilayer perceptron classifiers.

The authors apply oversampling and filter-based feature selection to improve classification quality and reduce the number of variables that must be tracked in real time. The best purchase-intention results are obtained with an MLP trained with resilient backpropagation and weight backtracking. For abandonment prediction, the study trains an LSTM-based recurrent neural network on sequential clickstream events. The two modules are then combined so that a site can identify visitors who appear willing to buy but are also likely to leave within the chosen prediction horizon.

Keywords: online shopping behavior, cart abandonment, clickstream data, deep learning.

1 Introduction

E-commerce activity has grown quickly, yet conversion rates have not always grown at the same pace. This creates a practical need for systems that can present timely and personalized interventions to online shoppers. In a physical store, sales staff can use experience to interpret shopper behavior and offer alternatives. Online services need data-driven methods that can play a similar role in a virtual environment.

Prior work has studied shopper navigation patterns, visit categorization, recommender systems, and real-time behavior prediction. Some studies group sessions according to clickstream patterns, while others try to infer future actions and reduce cart abandonment. This paper connects both ideas: it predicts buying intent and the likelihood of leaving the site at the same time.

The proposed system uses a purchase-intention module only when the abandonment module indicates that the visitor may leave soon. This design is meant to avoid unnecessary interventions and focus content offers on visitors who both show buying interest and face a near-term risk of leaving the site.

Earlier studies used clustering, association rules, support vector machines, k-nearest neighbors, hidden Markov models, and other machine-learning methods for related e-commerce tasks. This paper differs by combining session-level purchase-intention modeling with sequence-based abandonment prediction using LSTM recurrent neural networks.

2 Predicting Online Purchasing Intention

The purchase-intention problem is framed as binary classification. Each session is labeled according to whether the visit ended with a transaction. The dataset contains 12,330 sessions. Among these, 10,422 sessions are negative examples and 1,908 sessions are positive examples. The design attempts to avoid bias toward a particular user, campaign, special day, or time period by using sessions from different users over a one-year interval.

The feature set contains numerical variables describing page categories and time spent, Google Analytics-style measures such as bounce rate and exit rate, and categorical attributes such as browser, traffic source, visitor type, and month. The features are updated as the visitor moves through the website.

Table 1. Numerical features used in the behavior analysis model

Feature	Meaning	Min.	Max.	SD
Administrative	Number of account-management pages visited	0	27	3.32
Administrative duration	Time spent on account-management pages	0	3398	176.70
Informational	Number of informational pages visited	0	24	1.26
Informational duration	Time spent on informational pages	0	2549	140.64
Product related	Number of product-related pages visited	0	705	44.45
Product related duration	Time spent on product-related pages	0	63,973	1912.25
Bounce rate	Average bounce-rate value of visited pages	0	0.2	0.04
Exit rate	Average exit-rate value of visited pages	0	0.2	0.05
Page value	Average page-value score of visited pages	0	361	18.55
Special day	Distance of the visit date from a special shopping day	0	1.0	0.19

Table 2. Categorical features used in the behavior analysis model

Feature	Meaning	Categories
OperatingSystems	Visitor operating system	8

Feature	Meaning	Categories
Browser	Visitor browser	13
Region	Geographic region where the session began	9
TrafficType	Source that brought the visitor to the site	20
VisitorType	New visitor, returning visitor, or other	3
Weekend	Whether the visit happened on a weekend	2
Month	Month of the visit	12
Revenue	Class label showing whether the visit ended in a transaction	2

2.1 Classification Models

The study excludes lazy-learning methods such as k-nearest neighbors because the intended system is real-time and must be updated with new observations. The main comparison includes multilayer perceptron, decision-tree methods, and support vector machines. Categorical variables are encoded using 1-of-C coding, and numerical variables are standardized to zero mean and unit variance.

The MLP model has an input layer, one hidden layer, and an output layer. It uses nonlinear hidden-layer activation and a sigmoid output for binary classification. The network is trained using resilient backpropagation with weight backtracking, which gives each weight its own adaptive update scale.

The SVM models use both linear and radial basis function kernels. Hyperparameters are tuned with grid search on one selected train-validation split, and the resulting settings are reused for the repeated experiments. The decision-tree family includes C4.5 and random forest. The random forest uses bagging, random feature selection at each node, and 100 trees.

3 Filter-Based Feature Selection

Feature selection is used to improve performance and scalability. A real-time system benefits when it can track fewer variables without losing predictive power. Instead of applying feature extraction methods such as PCA, the paper uses filter-based feature selection so that selected variables remain interpretable and can be updated directly during a session.

The filters are correlation, mutual information, and mRMR. Mutual information can capture nonlinear relationships, while correlation focuses on linear association. The mRMR approach favors variables that are relevant to the class label but not redundant with variables already selected.

4 Predicting Likelihood of Abandonment

The abandonment module estimates whether a user will leave the site within a prediction horizon. The recurrent-neural-network dataset contains 185,000 page visits from 9,800 sessions and 3,500 visitors. Each event is represented by page type and time spent on the page. Product views, administrative actions, information-acquisition actions, and shopping-cart actions are represented as action categories.

The LSTM-RNN processes this sequence of page-view actions. Traditional RNNs can use prior temporal information, but they often struggle with long-range dependencies because of vanishing gradients. LSTM units address this issue with memory cells and gates controlling input, forgetting, and output exposure.

The prediction horizon is defined as the number of page-view actions before the visitor leaves. Based on earlier work, the model uses a step size of three. The LSTM-RNN has one hidden layer with 30 neurons. For comparison, the study also evaluates MLP, extreme learning machine, and radial basis function networks with a time-delay representation of recent actions.

5 Proposed System

The proposed workflow begins when the visitor reaches the landing page. Session and profile information are retrieved, and feature values are updated after each page view. The updated clickstream vector is sent to the LSTM-RNN abandonment model. If the abandonment probability exceeds its threshold, the purchase-intention MLP is evaluated. If the purchase-intention score also exceeds the selected threshold, the system displays content intended to keep the visitor engaged and encourage transaction completion.

Algorithm 1. User behavior analysis procedure

Step	Action
Input	Initial clickstream data X1, session data X2, abandonment threshold a1, purchase threshold a2
1	Repeat for each visitor action.
2	Update the clickstream representation.
3	Standardize the clickstream variables.
4	Use LSTM-RNN to estimate abandonment likelihood.
5	If abandonment likelihood exceeds a1, evaluate purchase intention.
6	Use MLP on clickstream and session features.
7	If purchase intention exceeds a2, display content.
8	Continue until the visitor ends the session.

6 Experimental Results

The purchase-intention experiments are repeated 100 times with random train-test partitions. The paper reports average accuracy, true-positive rate, true-negative rate, and F1 score. On the imbalanced dataset, high overall accuracy can be misleading because the negative class dominates. The classifiers tend to label many examples as the majority class.

After oversampling, the class distribution becomes balanced in the training data. This makes F1 score and true-positive rate more informative for identifying directed buying sessions. The strongest balanced-data result comes from the MLP with 10 hidden neurons.

Tables 3-5. Results on the imbalanced test set

Model	Accuracy (%)	TPR	TNR	F1
C4.5	88.92	0.57	0.96	0.57
Random forest	89.51	0.57	0.96	0.58
MLP, 10 hidden neurons	87.45	0.53	0.94	0.56
MLP, 20 hidden neurons	87.92	0.56	0.96	0.58
MLP, 40 hidden neurons	87.02	0.54	0.93	0.56
SVM, linear	88.25	0.42	0.97	0.52
SVM, RBF	86.14	0.46	0.92	0.53

Tables 6-9. Results after oversampling

Model	Accuracy (%)	TPR	TNR	F1
C4.5	82.34	0.79	0.85	0.82
Random forest	82.29	0.74	0.90	0.81
MLP, 10 hidden neurons	87.24	0.84	0.92	0.86
MLP, 20 hidden neurons	83.84	0.84	0.83	0.83
MLP, 40 hidden neurons	82.15	0.82	0.83	0.82
SVM, linear	84.26	0.75	0.93	0.82
SVM, RBF	84.88	0.75	0.94	0.82

Table 10. Feature rankings from the three filters

Rank	Correlation	Mutual information	mRMR
1	Page value	Page value	Page value
2	Exit rate	Exit rate	Month
3	Product related	Product related duration	Exit rate
4	Product related duration	Bounce rate	Weekend
5	Bounce rate	Product related	Informational duration
6	Administrative	Traffic type	Region
7	Visitor type	Administrative	Operating system
8	Informational	Month	Administrative
9	Administrative duration	Administrative duration	Visitor type
10	Special day	Informational	Product related duration

Table 11. Best MLP results from top-ranked features

Filter	Features	Accuracy (%)	TPR	TNR	F1
Correlation	9	84.32	0.84	0.85	0.84
Mutual information	10	84.11	0.84	0.85	0.84
mRMR	6	87.94	0.84	0.92	0.87

7 Abandonment Prediction Results

The LSTM-RNN outperforms the comparison neural-network models across prediction-horizon settings. The model predicts one-action-ahead abandonment with 74.3% accuracy. As the prediction horizon becomes longer, accuracy decreases because the final exit action is farther away and the user's future behavior is harder to estimate.

The purchase-intention module is triggered only after the abandonment score exceeds a threshold and at least three pages have been viewed. The highest true-positive rate reported for LSTM-RNN is 0.94 when the prediction horizon is 2 and the threshold is 0.90. Higher thresholds reduce the frequency with which the system suggests campaigns or content.

8 Conclusion

The study presents a real-time two-module system for online shopping behavior analysis. The first module predicts purchase intention from aggregated page-view, session, and user information. The second module predicts abandonment likelihood from sequential clickstream data. The modules are used together so that the system can focus interventions on users who seem likely to buy but also likely to leave soon.

For purchase-intention prediction, the strongest results are achieved by an MLP trained with resilient backpropagation and weight backtracking. Oversampling improves the handling of class imbalance, and mRMR feature selection gives the best compact model by selecting six highly useful variables. For abandonment prediction, the LSTM-RNN captures sequential behavior better than the other neural-network comparisons.

The findings suggest that clickstream statistics carry important buying-intent information, but session-level features provide additional nonredundant signal. A future extension could connect the prediction system to item-based or user-based recommendation models so that site-specific content can be offered to visitors who are both purchase-oriented and at risk of leaving.